Linear Algebra I, Sheet 1, MT2026

Systems of linear equations; matrices and their algebra.

Main course

1. Use any method you can think of to decide which (if any) of the following systems of linear equations with real coefficients have no solutions, which have a unique solution (in which case, what is it?), which have infinitely many solutions.

$$\text{(i)}\begin{cases} 2x+4y-3z &= 0\\ x-4y+3z &= 0\\ 3x-5y+2z &= 1 \end{cases}; \quad \text{(ii)}\begin{cases} x+2y+3z &= 0\\ 2x+3y+4z &= 1\\ 3x+4y+5z &= 2 \end{cases}; \quad \text{(iii)}\begin{cases} x+2y+3z &= 0\\ 2x+3y+4z &= 2\\ 3x+4y+5z &= 2 \end{cases}.$$

2. Let $A := \begin{pmatrix} 1 & 2\\ 3 & 4 \end{pmatrix}$, $B := \begin{pmatrix} 1\\ 2\\ 4 \end{pmatrix}$, $C := \begin{pmatrix} -1 & 5\\ 4 & 4 \end{pmatrix}$, $D := \begin{pmatrix} 1 & 4 & -3\\ 2 & 4 & -2 \end{pmatrix}$, $E := \begin{pmatrix} 1 & 2 \end{pmatrix}$ and $F := \begin{pmatrix} -1 & 5 & -6\\ 3 & 4 & -1 \end{pmatrix}$. For which pairs $X, Y \in \{A, B, C, D, E, F\}$ is $X - 2Y$ defined? And when it is defined, calculate it.

3. Calculate the following matrix products:

$$\begin{pmatrix} 1 & 2\\ 2 & 3 \end{pmatrix}\begin{pmatrix} x\\ y \end{pmatrix}; \quad \begin{pmatrix} 0 & 1\\ 2 & 3\\ 4 & 6 \end{pmatrix}\begin{pmatrix} 1 & 2\\ 2 & 3 \end{pmatrix}; \quad \begin{pmatrix} 1 & 2 & 3\\ 2 & 3 & 4\\ 3 & 4 & 5 \end{pmatrix}^2.$$

4. Let A be the 2×2 matrix $\begin{pmatrix} a & b\\ c & d \end{pmatrix}$.

(i) Show that A commutes with $\begin{pmatrix} 1 & 0\\ 0 & 0 \end{pmatrix}$ if and only if A is diagonal (that is, $b = c = 0$).

(ii) Which 2×2 matrices A commute with $\begin{pmatrix} 1 & 1\\ 0 & 0 \end{pmatrix}$?

(iii) Use the results of (i) and (ii) to find the matrices A that commute with every 2×2 matrix.

5. For each $\alpha \in \mathbb{R}$, find the RRE form for the matrix

$$\begin{pmatrix} 1 & 2 & -3 & -2 & 4 & 1\\ 2 & 5 & -8 & -1 & 6 & 2\\ 1 & 4 & -7 & 4 & 0 & \alpha \end{pmatrix}.$$

Use your result either to solve the following system of linear equations over $\mathbb{R}$, or to find values of α for which it has no solution:

$$\begin{cases} x_1 + 2x_2 - 3x_3 - 2x_4 + 4x_5 &= 1\\ 2x_1 + 5x_2 - 8x_3 - x_4 + 6x_5 &= 2\\ x_1 + 4x_2 - 7x_3 + 4x_4 &= \alpha \end{cases}$$

6. Let A and B denote square matrices with real entries. For each of the following assertions, find either a proof or a counterexample.

(i) $A^2 - B^2 = (A - B)(A + B)$.

(ii) If $AB = 0$ then $A = 0$ or $B = 0$.

(iii) If $AB = 0$ then A and B cannot both be invertible.

(iv) If A and B are invertible then $A + B$ is invertible.

(v) If $ABA = 0$ and B is invertible then $A^2 = 0$.

[Hint: where the assertions are false there are counterexamples of size 2×2.]

Starter

S1. Let $A := \begin{pmatrix} -1 & 3 \\ 1 & 2 \\ 7 & -2 \end{pmatrix}$, $B := \begin{pmatrix} 4 & -3 & 1 \\ 3 & -2 & 1 \\ 7 & 0 & 1 \end{pmatrix}$, $C := \begin{pmatrix} 2 & -2 & 3 \\ 0 & 0 & 0 \\ 5 & -4 & 3 \end{pmatrix}$. For which pairs X, $Y \in \{A, B, C\}$ is XY defined? When it is defined, calculate it.

S2. Calculate the following matrix products.

$$\begin{pmatrix} x & y & z & w \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix}; \quad \begin{pmatrix} x & y & z & w \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix}.$$

S3. Let A, B be invertible $n \times n$ matrices. Then AB is invertible, and $(AB)^{-1} = B^{-1}A^{-1}$.

Pudding

P1. Let $\alpha \in \mathbb{R}$ and

$$A = \begin{pmatrix} \cos\alpha & \sin\alpha \\ \sin\alpha & -\cos\alpha \end{pmatrix}.$$

(i) Show that $A^2 = I_2$.
(ii) Show that there is no real 2×2 matrix M such that $M^2 = A$.
(iii) Is there a matrix M with complex entries such that $M^2 = A$?

P2. We can identify a complex number $z = x + yi$ with the 2×2 real matrix Z by

$$x + yi = z \quad \leftrightarrow \quad Z = \begin{pmatrix} x & y \\ -y & x \end{pmatrix}.$$

Let z, w be two complex numbers such that $z \leftrightarrow Z$ and $w \leftrightarrow W$. Show that $z + w \leftrightarrow Z + W$ and $zw \leftrightarrow ZW$.

P3. (Here $\mathbb{Z}_n$ denotes the integers modulo n.) Put the following matrix into reduced row echelon form when working over (i) $\mathbb{R}$; (ii) $\mathbb{Z}_3$; (iii) $\mathbb{Z}_5$.

$$\begin{pmatrix} 1 & 1 & 1 & 0 & 1 \\ 1 & 2 & -1 & 2 & 0 \\ 0 & 1 & 1 & 2 & 1 \end{pmatrix}$$

How many solutions (x_1, x_2, x_3, x_4) are there to the linear system

$$x_1 + x_2 + x_3 = 1; \qquad x_1 + 2x_2 - x_3 + 2x_4 = 0; \qquad x_2 + x_3 + 2x_4 = 1;$$

when working over (i) $\mathbb{R}$; (ii) $\mathbb{Z}_3$ (iii) $\mathbb{Z}_5$?